

End Semester Examination 2023-24 even semester
CS 231: OPERATING SYSTEMS

MAX MARKS: 50

TIME 3HRS

NOTE: Attempt all questions. Question 1-14 is of 3 marks each and question-15 & 16 is of 4 marks each.

1. When the first fit for static allocation is beneficial?
2. What are the different parameters through which a memory allocation scheme is evaluated?
3. Which of the following addressing modes or combination of addressing modes are suitable for program relocation at run time? a. Absolute addressing b. Base addressing c. Relative addressing d. Indirect addressing
4. When the TLB implementation for reducing the memory access time fails?
5. What are the benefits of using virtual memory?
6. What is the reason for Belady's anomaly?
7. State the condition where the working set model for avoiding thrashing fails?
8. What are the desirable properties for a file system? What are the basic properties of a directory system?
9. List two in-memory and two on-disk data structure for any file system implementation.
10. State the best condition for Indexed Allocation of a file on the disk.
11. What is an "i-node"?
12. State the criteria/condition for choosing a particular RAID structure.
13. Consider a system with byte-addressable memory, 32-bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. What is the size of the page table in the system in megabytes?
14. Consider six memory partitions of sizes 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. If the best fit algorithm is used, which partitions are NOT allotted to any process?
15. A system uses FIFO policy for page replacement. It has 4 pages frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?
16. In a computer system where the 'best-fit' algorithm is used for allocating 'jobs' to 'memory partitions', the following situation was encountered:

Partitions sizes in KB	4K 8K 20K 2K
Jobs sizes in KB	2K 14K 3K 6K 6K 10K 20K 2K
Time for execution	4 10 2 1 4 1 8 6

When will the 20 K job complete?
